Exploring Urban Expansion in HCMC with Landsat Satellite Imagery: A Remote Sensing Analysis

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Introduction

Ho Chi Minh City (HCMC), the largest city in Vietnam, has undergone a dramatic urban transformation over the past few decades. Formerly known as Saigon, this economy-driving powerhouse has transformed from a war-scarred city into a thriving megacity with a population of over 9 million people as of 2025 (World Population Review, 2025). With this unprecedented urban expansion that is difficult to track with traditional monitoring methods, surrounding agricultural lands are being swallowed up at an alarming rate. As the city grows, consequences are seen in the environment with wetlands being lost to expansion, crucial agricultural land being developed, and sustainability challenges.

Remote sensing technology, particularly Landsat imagery from 1989-2024 shows that urban expansion in Ho Chi Minh City has accelerated, consuming vast amounts of agricultural lands and creating severe environmental impacts. This analysis highlights the critical role that remote sensing plays in monitoring and planning for sustainable urban development.

This paper will help to offer geographic context for HCMC's explosive growth through a remote sensing analysis, explain remote sensing methods for urban analysis, discuss findings from existing literature and analysis, outline potential environmental and socioeconomic impacts, and examine the implications for future urban development and sustainability.

Background & Context

HCMC is located in the Mekong Delta, the most productive agricultural region in Vietnam, responsible for 90% of Vietnam's rice exports, 70% of its fruit, and 60% of its seafood (Ministry of Agriculture and Environment). This city boasts an astounding population density of 4,097/km² and maintains itself as a key location for global trade and commerce.

Significant economic development in Vietnam has taken place since the 1986 Đổi Mới reforms which were responsible for creating a market economy in Vietnam. The country has also experienced a tremendous increase in Foreign Direct Investment (FDI). This influx of capital and shift to a market economy has driven HCMC's rapid growth and its 23% contribution to Vietnam's GDP (World Population Review, 2025).

The Mekong Delta contributes heavily to Vietnam's overall food production and is essential to food security. The delta is facing environmental challenges of saltwater intrusion, land subsidence, and vulnerability to climate change. The delta's low-lying topography makes it a flood-prone region, with risks further amplified by land development.

Remote Sensing Methods

Remote sensing offers key advantages over traditional ground surveys including the ability to track changes over time using consistent and repeatable measurements. With imagery readily available and wide-ranging in its applicability, remote sensing offers a unique value proposition of low cost of entry and excellent data for analyzing large areas. The historical database of imagery from Landsat beginning in 1972 enables the long-term analysis of HCMC. These advantages make remote sensing particularly well-suited for examining HCMC's urban development over time.

Landsat imagery, the basis for this analysis, originates from the Landsat program, launched in 1972. The program has evolved over time with multiple satellite generations being deployed. This analysis uses data from Landsat 5, 7, 8, and 9. Landsat satellites utilize both multispectral and panchromatic sensors with multispectral sensors having a spatial resolution of 30m and panchromatic 15m. The data archive is freely accessible to the public, with Landsat Collection 2 serving as the current processing standard.

Satellites detect urban areas by identifying distinct electromagnetic signatures. Spectral signatures vary based on land cover type and offer insight on the land being studied. Various multispectral bands contain different use cases that allow for specific analysis of different factors. The visible bands (RGB) are useful in creating true color visualizations, near-infrared (NIR) bands are used for vegetation detection, shortwave infrared (SWIR) is used for built-up area detection, and thermal bands are useful in monitoring temperature or heat detection. These bands can be used independently or in combination; band combinations help identify details that may be missed through traditional visual analysis.

Analysis of urban expansion in HCMC was conducted in Google Earth Engine using multi-temporal Landsat imagery from 1989 to 2024. Reflectance data from Landsat 5, 7, 8, and 9 Collection 2 were used for the five time periods (1989, 2000, 2010, 2020, and 2024). Cloud masking and median composites were generated for each time period to reduce atmospheric interference. Usage of three spectral indices: Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI), and Normalized Difference Water Index (NDWI) were required to effectively classify urban areas.

Urban area pixels were classified as pixels where NDVI < 0.2, NDBI > 0, and NDWI < -0.1, this method effectively identified urban areas while filtering out vegetation, water bodies, and saturated soils. The study was conducted at a 30-meter spatial resolution and resulted in urban area values in km². Analysis was performed to detect changes in urban areas between time periods.

Results

HCMC has undergone extraordinary urban expansion, growing 6.5x between 1989 and 2006, with continued acceleration through 2024. With this explosive growth comes major agricultural land conversion. Traditional characteristics of urban sprawl are shown by the urban area images. Expansion northwards and eastward of the city center exhibits peri-urbanization characteristics of forced development in areas closest to the city center. Rice paddies and other agricultural lands have rapidly disappeared as the city continues to grow.

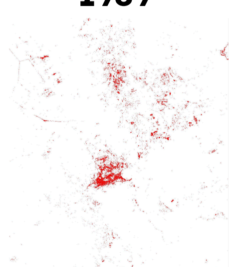
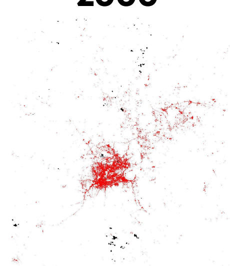
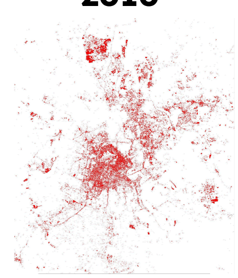
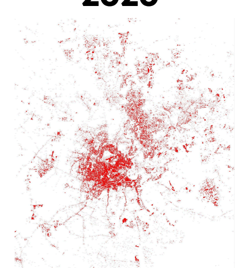
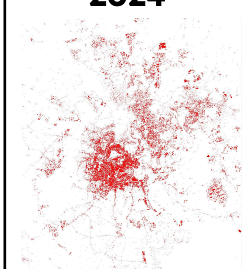
HCMC Urban Area in (km²)				
1989	**2000**	**2010**	**2020**	**2024**
				
82.93	97.53	265.61	272.09	281.75

Figure 1. Urban area classification maps of HCMC (1989-2024).

The Google Earth Engine spatial analysis revealed distinct urban expansion patterns in HCMC. The primary growth occurred northward along Highway 13, eastward towards the industrial districts of Thủ Đức and Bình Dương, and radially from the city center. HCMC's urban expansion shows core densification and peripheral sprawl development in the outer areas of the city. True color imagery of HCMC displays these expansion patterns.

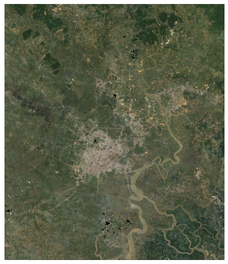
HCMC True Color				
1989	**2000**	**2010**	**2020**	**2024**
				

Figure 3. True color image series showing urban expansion of HCMC (1989-2024).

The true color image series illustrates the change in HCMC's landscape as agricultural lands are rapidly developed. Rapid losses of agricultural land shown 1989-2010 indicate peri-urbanization. The images highlight the densification of the historic city center with formerly green pockets of land turning into urban landscape over time. The periphery of the city shows leap-frog development with districts expanding around the edge of the city.

Ho Chi Minh City Urban Area Table		
Period	**Urban Area Change (km²)**	**Percent Increase**
1989 → 2000	82.93 → 97.53	**17.61%**
2000 → 2010	97.53 → 265.61	**172.33%**
2010 → 2020	265.61 → 272.09	**2.44%**
2020 → 2024	272.09 → 281.75	**3.55%**

Figure 2. Table displaying changes in HCMC urban area (1989-2024).

The table shown in Figure 2 indicates the change in HCMC urban area (km^2), illustrating the dramatic acceleration in urbanization during the 2000-2010 period (172% growth) and a plateau effect in the following decades as the city reached limits for easily developable areas. HCMC's rapid urbanization is defined by the 239.74% growth between 1989-2024.

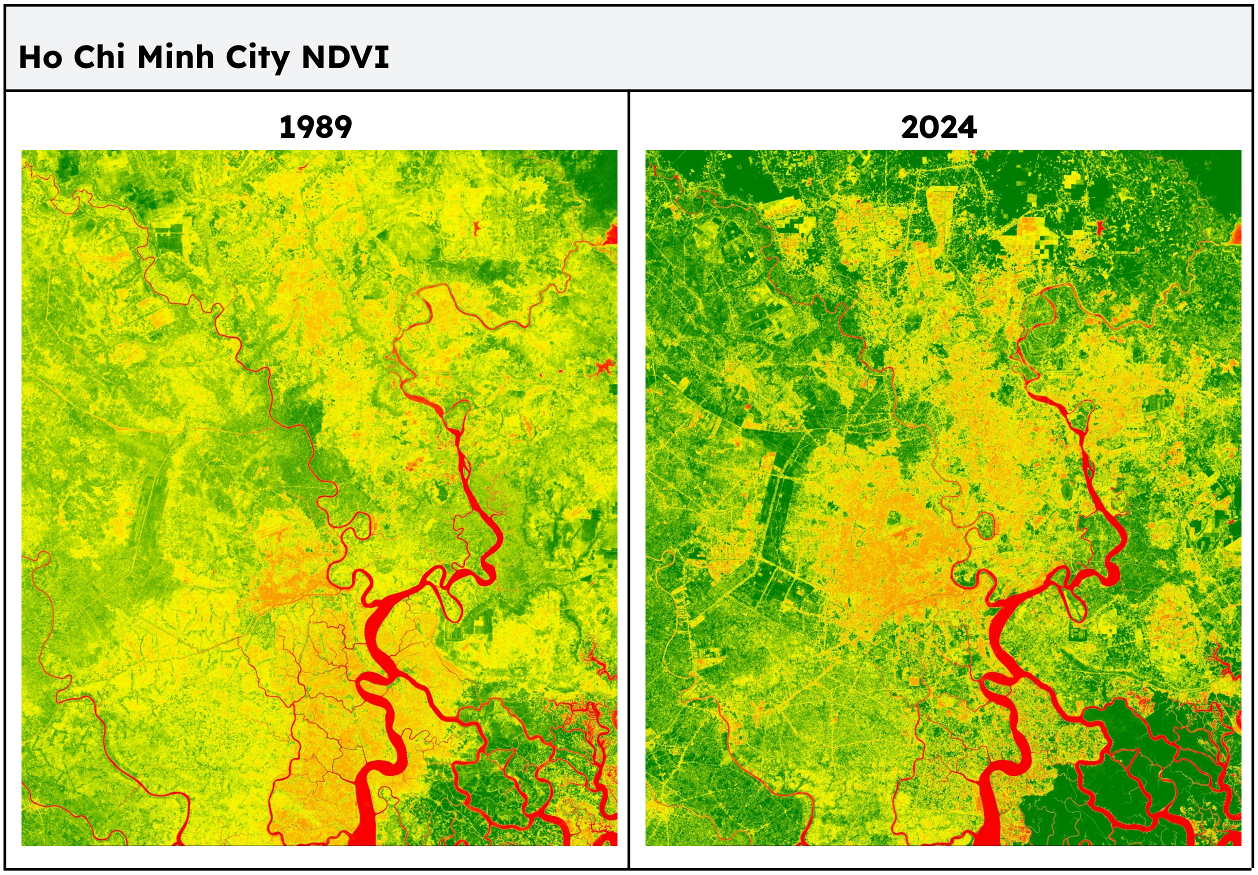


Figure 4. Normalized Difference Vegetation Index (NDVI) of HCMC (1989-2024).

NDVI comparison results show a transformation of surrounding agricultural rice paddies being converted into residential and industrial areas; this supports previous findings of up to a 66% loss of agricultural lands in northern districts. Some historical rice paddies are now completely urbanized.

Discussion

Rapid growth and development are outpacing infrastructure availability and capacity to support the expanding metropolis. The city's transportation network is crowded, faced with an influx of people. The city lacks adequate drainage and the climate's monsoon season leads to flooding all over the city. Development is often informal as in a city so large, the enforcement of rules and codes is impractical. Unregistered residents can be found throughout the city as a result of the social migration occurring. In addition to these challenges, districts are failing to coordinate and agricultural land falls victim to these failures with enforcement of land use regulations being difficult.

Urban Heat Island Effect (UHI) is a popular method for monitoring environmental changes. Research by Son et al. (2017) found that industrial zones exhibited surface temperatures exceeding 45°C, while urban residential areas ranged from 36-40°C and rural/vegetated areas remained at 30-35°C. These findings underscore a substantial 10-15°C temperature difference between the urban core and periphery, demonstrating the severe urban heat island effect. HCMC is also experiencing severe wetland losses in its southern districts. This decline in wetlands increases the risk of flooding and reduces the region's natural drainage capacity. These changes reflect in the declining water quality of the Saigon River (Nhu et al., 2018). While HCMC's economic development is fierce, its surrounding ecosystem is suffering, taking massive losses in agricultural productivity, facing microclimate changes, and wetland losses leading to decreased biodiversity, increased flood risks, and declining water quality.

Urban development in HCMC has resulted in significant employment shifts, with approximately 13 jobs lost per hectare of converted agricultural land (Schaefer & Thinh, 2019). In addition to these losses, many farming communities are being displaced as the city shifts from an agricultural to industrial economy. The region exhibits typical rural-to-urban migration patterns as farmers, facing displacement from agricultural land conversion, seek economic opportunities in the city. Students leave their home towns to study in the bustling HCMC. With these changes and a growing city population comes an increased food burden and major food security implications for the entire region.

Remote sensing offers a new method for monitoring and tracking changes through continuous satellite monitoring. Findings and analyses from satellite imagery can be utilized to assist urban planning processes and evaluate decisions. Real-time data offers planners and researchers capability to make more informed decisions while an immense history of imagery enables more complex analysis of cause-effect relationships and change over time. When used in conjunction with ground surveys and ground-reference data, multi-scale analyses offer a wealth of data for decision makers on both a regional and local scale. Data-driven evidence provides policymakers with objective information to inform improved policies, identify high-priority areas in need of intervention, and monitor the effectiveness of urban planning decisions.

Conclusion

Analysis using remote sensing methods provides objective and comprehensive monitoring of HCMC through long-term analysis that is impossible with traditional methods. Utilization of Landsat's freely available imagery archive democratizes research capabilities, while tools like Google Earth Engine make sophisticated analysis accessible to researchers worldwide. Remote sensing is a critical tool for planners helping to illustrate complex urbanization dynamics and enabling evidence-based policy making. Its capabilities support sustainable urban planning and reliable long-term monitoring.

HCMC is a valuable case study for other Southeast Asian countries on rapid urbanization as we see similar patterns of development taking place in cities like Bangkok, Jakarta, and Manila. The methods and insights used can be leveraged to help planners mitigate risks and develop balanced approaches between development and sustainability. Remote sensing is an essential tool for urban management in the 21st century and truly democratizes data analysis and monitoring capabilities.

Future Research

HCMC will likely continue to develop as Vietnam continues to grow economically. This will lead to increased pressure on remaining agricultural lands and intensified climate challenges. There is an opportunity for comprehensive approaches that combine remote sensing with ground-based monitoring to help shift urban planning from a reactive to proactive approach.

For sustainable development, the city must balance economic growth with environmental protection, particularly preservation of critical agricultural lands.

Final Thought

Ho Chi Minh City offers many valuable insights that are applicable to other regions undergoing rapid urbanization. While urban populations continue to expand, remote sensing will continue to be a powerful tool for monitoring and understanding sustainable urban development. The capabilities offered through satellite imagery archives and cloud-based analysis platforms offer planners new hope for creating livable and sustainable cities.

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